

AP CALCULUS AB – UNIT 2

2.4 Basic Derivative Rules



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Differentiate powers with positive, negative, and rational exponents.
- Apply the constant, sum, difference, and constant-multiple rules accurately.
- Differentiate $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$.
- Differentiate e^x and $\ln x$ on their appropriate domains.
- Combine foundational derivative rules in the correct order.
- Simplify expressions strategically before or after differentiating.
- Use derivative formulas to determine slopes, tangent lines, and rates of change.

Keys: Essential Knowledge and Vocabulary

Power Rule: Power Rule	$\frac{d}{dx}x^n = nx^{n-1}$ wherever the original function and derivative are defined.
Power Rule: Radical notation	$\sqrt[q]{x^p} = x^{p/q}$ on the appropriate real domain.
Power Rule: Negative exponent	$x^{-m} = 1/x^m$ for $x \neq 0$.
Power Rule: Domain check	A derivative formula does not erase restrictions inherited from the original function.
Linearity Rules: Constant Rule	$\frac{d}{dx}(C) = 0$.
Linearity Rules: Constant Multiple Rule	$\frac{d}{dx}[cf(x)] = cf'(x)$.
Linearity Rules: Sum/Difference Rules	$(f \pm g)' = f' \pm g'$.
Linearity Rules: Linearity	$D[af + bg] = aDf + bDg$ for constants a, b .
Core Transcendental Derivatives: Sine and cosine	$(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$ when x is measured in radians.
Core Transcendental Derivatives: Natural exponential	$(e^x)' = e^x$.
Core Transcendental Derivatives: Natural logarithm	$(\ln x)' = 1/x$ for $x > 0$.
Core Transcendental Derivatives: Unit-circle discipline	Use exact trigonometric values whenever the angle is standard and no calculator is allowed.
Tangent, Cotangent, Secant, and Cosecant: Tangent	$(\tan x)' = \sec^2 x$.
Tangent, Cotangent, Secant, and Cosecant: Cotangent	$(\cot x)' = -\csc^2 x$.
Tangent, Cotangent, Secant, and Cosecant: Secant	$(\sec x)' = \sec x \tan x$.
Tangent, Cotangent, Secant, and Cosecant: Cosecant	$(\csc x)' = -\csc x \cot x$.

Explanation, Strategy, and Common Errors

- **Power Rule.** Lower the exponent by one after multiplying by the original exponent. Keep exact fractional exponents unless the problem requests radical form.
- **Power Rule.** Check where the function is real and defined. For example, $x^{1/2}$ is real for $x \geq 0$, while its derivative $1/(2\sqrt{x})$ requires $x > 0$.
- **Power Rule.** A limit that matches the derivative definition of a power function can often be evaluated immediately by the Power Rule.
- **Linearity Rules.** Differentiation is linear: constants can be pulled through the derivative, and sums can be differentiated term by term.
- **Linearity Rules.** First simplify algebraically when it reduces work, especially when common factors or quotients of powers can be rewritten. Do not invent a rule such as $(fg)' = f'g'$; products require the Product Rule in Topic 2.8.
- **Linearity Rules.** After differentiating, evaluate only if a particular input is requested. Keeping symbolic structure often reduces arithmetic errors.
- **Core Transcendental Derivatives.** The standard trigonometric derivative formulas assume radians. Degree measure introduces an additional conversion factor.
- **Core Transcendental Derivatives.** For sums and constant multiples, differentiate each term separately. Composite functions are postponed to the Chain Rule in Unit 3.
- **Core Transcendental Derivatives.** Preserve the domain of $\ln x$. A derivative formula evaluated outside the original function domain is meaningless.
- **Tangent, Cotangent, Secant, and Cosecant.** Keep signs attached to cotangent and cosecant derivative rules. A compact memory pattern is: tangent and secant are positive; cotangent and cosecant are negative.
- **Tangent, Cotangent, Secant, and Cosecant.** Use unit-circle values to evaluate exactly. Check that the original function is defined at the requested angle before evaluating its derivative.
- **Tangent, Cotangent, Secant, and Cosecant.** These formulas can be derived from $\tan = \sin/\cos$, $\cot = \cos/\sin$, $\sec = 1/\cos$, and $\csc = 1/\sin$ using earlier rules.

Solved Example: Power Rule - Polynomial power

Differentiate $f(x) = x^7$.

Answer and Reasoning

$$f'(x) = 7x^6.$$

Solved Example: Power Rule - Radical and reciprocal

Differentiate $g(x) = \sqrt[3]{x^5} + \frac{1}{x^4}$.

Answer and Reasoning

$$g(x) = x^{5/3} + x^{-4}, \text{ so } g'(x) = \frac{5}{3}x^{2/3} - 4x^{-5}.$$

Solved Example: Power Rule - Derivative-definition limit

Evaluate $\lim_{h \rightarrow 0} \frac{(3+h)^8 - 3^8}{h}$.

Answer and Reasoning

This is the derivative of x^8 at 3, so the value is $8 \cdot 3^7$.

Solved Example: Linearity Rules - Polynomial

Differentiate $f(x) = 7x^5 - 3x^2 + 9x - 4$.

Answer and Reasoning

$$f'(x) = 35x^4 - 6x + 9.$$

Solved Example: Linearity Rules - Fractional powers

Differentiate $g(x) = 4x^{3/2} - 6x^{-1/3} + 11$.

Answer and Reasoning

$$g'(x) = 6x^{1/2} + 2x^{-4/3}.$$

Solved Example: Linearity Rules - Linearity from known derivatives

If $f'(2) = 5$ and $g'(2) = -1$, find the derivative of $H(x) = 3f(x) - 4g(x) + 9$ at 2.

Answer and Reasoning

$$H'(2) = 3(5) - 4(-1) = 19.$$

Solved Example: Core Transcendental Derivatives - Mixed basic derivatives

Differentiate $f(x) = 4 \sin x - 3 \cos x + 2e^x - 5 \ln x$.

Answer and Reasoning

$$f'(x) = 4 \cos x + 3 \sin x + 2e^x - \frac{5}{x}.$$

Solved Example: Core Transcendental Derivatives - Exact evaluation

Find $g'(\pi/3)$ for $g(x) = 2 \sin x + \cos x$.

Answer and Reasoning

$$g' = 2 \cos x - \sin x, \text{ so } g'(\pi/3) = 1 - \frac{\sqrt{3}}{2}.$$

Solved Example: Core Transcendental Derivatives - Tangent line to a logarithm

Write the tangent line to $y = \ln x$ at $x = e$.

Answer and Reasoning

$$\text{Point } (e, 1), \text{ slope } 1/e; y - 1 = \frac{1}{e}(x - e).$$

Solved Example: Tangent, Cotangent, Secant, and Cosecant - Basic derivatives

Differentiate $f(x) = 3 \tan x - 2 \csc x$.

Answer and Reasoning

$$f'(x) = 3 \sec^2 x + 2 \csc x \cot x.$$

Solved Example: Tangent, Cotangent, Secant, and Cosecant - Exact angle

Find $(\sec x)'$ at $x = \pi/3$.

Answer and Reasoning

$$\sec(\pi/3) \tan(\pi/3) = 2\sqrt{3}.$$

Solved Example: Tangent, Cotangent, Secant, and Cosecant - Product with a trig function

Differentiate $g(x) = x^2 \cot x$.

Answer and Reasoning

$$g'(x) = 2x \cot x - x^2 \csc^2 x.$$

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. Power Rule: Differentiate x^9 .

2. Power Rule: Differentiate x^{-3} .

3. Power Rule: Differentiate $\sqrt{x}$.

4. Power Rule: Differentiate $\sqrt[4]{x^3}$.

5. Power Rule: Find the slope of $y = x^5$ at $x = -1$.

6. Linearity Rules: Differentiate $8x^4 - 5x + 12$.

7. Linearity Rules: Differentiate $3\sqrt{x} + 7/x$.

8. Linearity Rules: If $f'(a) = 2$, find $(5f - 9)'(a)$.

9. Linearity Rules: Differentiate $\pi x^3 - e^2$.

10. Linearity Rules: Find the derivative of $x^6 - x^6 + 4$.

11. Core Transcendental Derivatives: Differentiate $\sin x$.

12. Core Transcendental Derivatives: Differentiate $7 \cos x$.

13. Core Transcendental Derivatives: Differentiate $3e^x - 4$.

14. Core Transcendental Derivatives: Differentiate $5 \ln x$.

15. Core Transcendental Derivatives: Find $(\sin x)'$ at $x = 0$.

16. Tangent, Cotangent, Secant, and Cosecant: Differentiate $\tan x$.

17. Tangent, Cotangent, Secant, and Cosecant: Differentiate $5 \cot x$.

18. Tangent, Cotangent, Secant, and Cosecant: Differentiate $-3 \sec x$.

19. Tangent, Cotangent, Secant, and Cosecant: Differentiate $2 \csc x$.

20. Tangent, Cotangent, Secant, and Cosecant: Find $(\tan x)'$ at $x = 0$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

21. Power Rule: Differentiate $f(x) = x^{7/3} - x^{-2}$.

22. Power Rule: Find $f'(16)$ for $f(x) = x^{3/4}$.

23. Power Rule: Write the tangent line to $y = x^{3/2}$ at $x = 4$.

24. Power Rule: Evaluate $\lim_{x \rightarrow 2} \frac{x^{10} - 1024}{x - 2}$.

25. Power Rule: For $f(x) = x^{-1/2}$, state the real domain of f and f' .

26. Linearity Rules: Differentiate $f(x) = \frac{2x^5 - 3x^2 + 1}{x^2}$ after rewriting.

27. Linearity Rules: If $f'(1) = -3$ and $g'(1) = 4$, find $H'(1)$ for $H = 2f + \frac{1}{2}g$.

28. Linearity Rules: Find the tangent line to $y = 2x^3 - 5x^2 + 4$ at $x = 1$.

29. Linearity Rules: Find all critical slopes equal to zero for $f(x) = x^3 - 3x$.

30. Linearity Rules: Evaluate $\lim_{h \rightarrow 0} \frac{[3(2+h)^4 - 5(2+h)] - [3(2)^4 - 10]}{h}$.

31. Core Transcendental Derivatives: Differentiate $f(x) = x^4 + 2 \sin x - e^x$.

32. Core Transcendental Derivatives: Find the tangent line to $y = e^x$ at $x = 0$.

33. Core Transcendental Derivatives: Find $f'(\pi/2)$ for $f(x) = 3 \sin x + 4 \cos x$.

34. Core Transcendental Derivatives: Find all x in $[0, 2\pi]$ where the derivative of $\sin x + \cos x$ is zero.

35. Core Transcendental Derivatives: Find the slope of $y = \ln x + x^{-1}$ at $x = 1$.

36. Tangent, Cotangent, Secant, and Cosecant: Find $(\cot x)'$ at $x = \pi/4$.

37. Tangent, Cotangent, Secant, and Cosecant: Find $(\csc x)'$ at $x = \pi/6$.

38. Tangent, Cotangent, Secant, and Cosecant: Differentiate $f(x) = x \sec x$.

39. Tangent, Cotangent, Secant, and Cosecant: Differentiate $g(x) = \frac{\tan x}{x}$ for $x \neq 0$.

40. Tangent, Cotangent, Secant, and Cosecant: State where $f(x) = \sec x + \csc x$ is defined.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

41. Power Rule: Differentiate $f(x) = \frac{x^5+1}{x^3}$ after rewriting.

42. Power Rule: Find all points on $y = x^{4/3}$ where the tangent slope is 4.

43. Power Rule: Find the tangent line to $y = \sqrt[3]{x}$ at $x = 8$.

44. Power Rule: Determine where $f(x) = x^{2/3}$ has a horizontal tangent and where it is nondifferentiable.

45. Linearity Rules: Find a, b if $f(x) = ax^3 + bx^2 - 5x$ has $f'(1) = 4$ and $f'(2) = 19$.

46. Linearity Rules: If $F(x) = af(x) + bg(x)$ and $F'(x) = 0$ for all x , what relation holds between f' and g' ?

47. Linearity Rules: Find a polynomial p of least degree satisfying $p'(x) = 6x^2 - 4x + 3$ and $p(0) = 7$.

48. Linearity Rules: Find the equation of the normal line to $y = x^4 - 2x$ at $x = 1$.

49. Core Transcendental Derivatives: Find the normal line to $y = 2 \sin x$ at $x = \pi/3$.

50. Core Transcendental Derivatives: Find all $x > 0$ where $f(x) = e^x - \ln x$ has derivative zero.

51. Core Transcendental Derivatives: Evaluate $\lim_{h \rightarrow 0} \frac{\sin(\pi/6+h) - 1/2}{h}$.

52. Core Transcendental Derivatives: Evaluate $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$.

53. Tangent, Cotangent, Secant, and Cosecant: Find the tangent line to $y = \tan x$ at $x = \pi/4$.

54. Tangent, Cotangent, Secant, and Cosecant: Find all $x \in [0, 2\pi]$ where the derivative of $\sec x$ is zero.

55. Tangent, Cotangent, Secant, and Cosecant: Derive $(\tan x)' = \sec^2 x$ from the Quotient Rule.

56. Tangent, Cotangent, Secant, and Cosecant: Differentiate $f(x) = \sec x \tan x$ using the Product Rule.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

57. Power Rule: For $f(x) = x^p$ with $p \neq 0$, solve $f'(1) = f(1)$.

58. Power Rule: Find n if the tangent slope of $y = x^n$ at $x = 2$ is 48 and n is a positive integer.

59. Power Rule: Determine all real $x > 0$ where the tangent to $y = x^{5/2}$ is parallel to the secant from $x = 0$ to $x = 4$.

60. Linearity Rules: For $f(x) = ax^4 + bx^3 + cx^2$, find conditions for $f'(1) = 0$ and $f'(-1) = 0$.

61. Linearity Rules: Suppose p is a polynomial and $p'(x) = p'(-x)$ for all x . What can be said about the nonconstant terms of p ?

62. Linearity Rules: Find all k for which the line $y = kx + 1$ is tangent to $f(x) = x^2 - 3x + 4$.

63. Core Transcendental Derivatives: Find constants a, b so $f(x) = a \sin x + b \cos x$ satisfies $f(0) = 2$ and $f'(0) = -3$.

64. Core Transcendental Derivatives: For $f(x) = e^x - x$, show that the tangent at $x = 0$ is horizontal and write it.

65. Core Transcendental Derivatives: Find all points on $y = \ln x$ whose tangent line passes through the origin.

66. Tangent, Cotangent, Secant, and Cosecant: Find all $x \in (0, \pi)$ where $f(x) = \tan x - x$ has a horizontal tangent and is defined.

67. Tangent, Cotangent, Secant, and Cosecant: Find the normal line to $y = \csc x$ at $x = \pi/2$.

68. Tangent, Cotangent, Secant, and Cosecant: Show that $(\sec x + \tan x)' = \sec x(\sec x + \tan x)$.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

69. Power Rule: Using only the Power Rule, show that the derivative of $x^{m/n}$ is $\frac{m}{n}x^{m/n-1}$ for positive x by differentiating $y^n = x^m$ informally and solving for y' .

70. Power Rule: For which $p > 0$ is x^p differentiable at 0 with derivative 0 on a right-hand domain?

71. Linearity Rules: Prove from the derivative definition that $(cf)'(a) = cf'(a)$ for constant c .

72. Linearity Rules: If $f'(x) = 0$ for every x in an interval, what conclusion is expected, and which later theorem formally proves it?

73. Core Transcendental Derivatives: Explain why the derivative formulas for sine and cosine depend on radian measure.

74. Core Transcendental Derivatives: Find A, B such that $A \sin x + B \cos x$ has value 5 and derivative 0 at $x = \pi/6$.

75. Tangent, Cotangent, Secant, and Cosecant: Find constants A, B so $F(x) = A \sec x + B \tan x$ satisfies $F(0) = 2$ and $F'(0) = 3$.

76. Tangent, Cotangent, Secant, and Cosecant: Prove $(\csc x)' = -\csc x \cot x$ from the reciprocal form $1/\sin x$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. **Power Rule:** $\frac{d}{dx}x^{-5}$ equals

- (A) $-5x^{-6}$ (B) $-5x^{-4}$
(C) x^{-6} (D) $5x^4$

2. **Power Rule:** The derivative of $\sqrt[3]{x^2}$ for $x \neq 0$ is

- (A) $\frac{2}{3}x^{-1/3}$ (B) $\frac{3}{2}x^{1/3}$
(C) $2x^{-2/3}$ (D) $\frac{1}{3}x^{-2/3}$

3. **Power Rule:** $\lim_{h \rightarrow 0} \frac{(5+h)^6 - 5^6}{h}$ equals

- (A) 5^6 (B) $6 \cdot 5^5$
(C) $6 \cdot 5^6$ (D) $5 \cdot 6^5$

4. **Power Rule:** At which $x > 0$ does $y = \sqrt{x}$ have slope $\frac{1}{8}$?

- (A) 2 (B) 4
(C) 8 (D) 16

5. **Linearity Rules:** If $H = 4f - 3g + 7$, then $H' =$

- (A) $4f' - 3g'$ (B) $4f'g'$
(C) $4f - 3g$ (D) $f' - g' + 7$

6. **Linearity Rules:** The derivative of $5x^{3/2} - 2x^{-1} + 8$ is

- (A) $\frac{15}{2}x^{1/2} + 2x^{-2}$ (B) $\frac{3}{2}x^{1/2} - x^{-2}$
(C) $15x^{1/2} + 2x^{-1}$ (D) $\frac{15}{2}x^{3/2} + 2x^{-2}$

7. **Linearity Rules:** If $f'(2) = 3$ and $g'(2) = 5$, then $(2f - g)'(2) =$

- (A) 1 (B) 6
(C) 11 (D) -1

8. **Linearity Rules:** Which simplification is most useful before differentiating $\frac{x^4 + x}{x}$?

- (A) $x^3 + 1$ (B) $x^4 + 1$
(C) $x^3 + x$ (D) x^5

9. **Core Transcendental Derivatives:** $\frac{d}{dx}(3 \cos x - 2 \sin x)$ is

- (A) $-3 \sin x - 2 \cos x$ (B) $3 \sin x - 2 \cos x$
(C) $3 \cos x + 2 \sin x$ (D) $-3 \cos x - 2 \sin x$

10. **Core Transcendental Derivatives:** The derivative of $e^x + \ln x$ is

- (A) $xe^{x-1} + 1/x$ (B) $e^x + x$
(C) $e^x + 1/x$ (D) $e^{x-1} + \ln x$

11. Core Transcendental Derivatives: At $x = \pi$, the slope of $y = \sin x$ is

- (A) -1 (B) 0
(C) 1 (D) π

12. Core Transcendental Derivatives: $\lim_{h \rightarrow 0} \frac{e^{2+h} - e^2}{h}$ equals

- (A) $2e$ (B) e^2
(C) $2e^2$ (D) $1/e^2$

13. Tangent, Cotangent, Secant, and Cosecant: $(\cot x)' =$

- (A) $\csc^2 x$ (B) $-\csc^2 x$
(C) $\sec^2 x$ (D) $-\sec^2 x$

14. Tangent, Cotangent, Secant, and Cosecant: $(\sec x)' =$

- (A) $\sec^2 x$ (B) $\sec x \tan x$
(C) $-\sec x \tan x$ (D) $\csc x \cot x$

15. Tangent, Cotangent, Secant, and Cosecant: At $x = \pi/4$, $(\tan x)'$ is

- (A) 1 (B) $\sqrt{2}$
(C) 2 (D) 4

16. Tangent, Cotangent, Secant, and Cosecant: Which derivative is negative at $x = \pi/4$?

- (A) $(\tan x)'$ (B) $(\sec x)'$
(C) $(\cot x)'$ (D) All are positive.

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

17. Power Rule: A calculator evaluation of $f'(7.2)$ for $f(x) = x^{1.7}$ is closest to

- (A) 1.7 (B) 4.01
(C) 6.77 (D) 28.7

18. Power Rule: For $g(x) = x^{-2.4}$, $g'(1.3)$ is closest to

- (A) -2.40 (B) -0.984
(C) 0.984 (D) 2.40

19. Linearity Rules: For $p(x) = 1.7x^{4.2} - 3.1x + 8$, $p'(2)$ is closest to

- (A) 7.14 (B) 48.80
(C) 62.51 (D) 101.61

20. Linearity Rules: A model $R(t) = 1200 + 85t - 4.7t^2$ has instantaneous rate at $t = 6$ closest to

- (A) 28.6 (B) 56.4
(C) 85.0 (D) 113.2

21. Core Transcendental Derivatives: For $f(x) = e^x - 4 \ln x$, $f'(2.3)$ is closest to

- | | |
|----------|-----------|
| (A) 0.31 | (B) 8.23 |
| (C) 9.97 | (D) 11.71 |

22. Core Transcendental Derivatives: The positive solution of $e^x = 1/x$ is closest to

- | | |
|-----------|-----------|
| (A) 0.368 | (B) 0.567 |
| (C) 1.000 | (D) 1.763 |

23. Tangent, Cotangent, Secant, and Cosecant: At $x = 0.7$, $(\sec x)'$ is closest to

- | | |
|----------|----------|
| (A) 0.84 | (B) 1.10 |
| (C) 1.43 | (D) 2.18 |

24. Tangent, Cotangent, Secant, and Cosecant: At $x = 1.1$, the derivative of $3 \csc x - 2 \tan x$ is closest to

- | | |
|------------|-----------|
| (A) -11.43 | (B) -8.35 |
| (C) -4.15 | (D) 11.43 |

AP-Style Free-Response Practice

Calculator Permitted

Complete the following independent calculator-permitted problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Power Rule. Let $f(x) = x^{5/3} - 3x^{-1/2}$.

Find $f'(x)$ and state its real domain.

[3 points]

(1b) Find the tangent slope at $x = 1$.

[1 points]

(1c) Write the tangent line at $x = 1$.

[2 points]

(2a) Problem 2: Linearity Rules. Let $p(x) = ax^3 + bx^2 + cx + d$ satisfy $p(0) = 2$, $p'(0) = -1$, $p'(1) = 4$, and $p'(2) = 15$.

Use the first two conditions to determine c and d .

[2 points]

(2b) Form equations for a and b from the remaining derivative conditions.

[2 points]

(2c) Solve for a , b and state $p(x)$.

[3 points]

(3a) Problem 3: Core Transcendental Derivatives. Let $f(x) = 2 \sin x + 3 \cos x + \ln x$ for $x > 0$.

Find $f'(x)$.

[2 points]

(3b) Find the exact tangent slope at $x = \pi/2$. [2 points]

(3c) Write the tangent-line equation at $x = \pi/2$. [3 points]

(4a) **Problem 4: Tangent, Cotangent, Secant, and Cosecant.** Let $f(x) = 2 \sec x - \tan x$ on an interval where it is defined.

Find $f'(x)$. [2 points]

(4b) Find the exact value of $f'(\pi/3)$. [2 points]

(4c) Write the tangent line at $x = \pi/3$. [3 points]

No Calculator

Complete the following independent non-calculator problems. Show the mathematical setup and justify conclusions when requested.

(1a) **Problem 1: Power Rule.** A model is $P(t) = 84t^{0.62}$ for $t > 0$.

Use a calculator to find $P'(5)$ to three decimals. [2 points]

(1b) Interpret the value if P is measured in grams and t in days. [2 points]

(1c) Compare $P'(5)$ and $P'(20)$ and explain the trend.

[2 points]

(2a) Problem 2: Linearity Rules. The number of users of a service is modeled by $U(t) = 1250 + 480t^{1.35} - 22t^2$, where t is months.

Find an expression for $U'(t)$.

[2 points]

(2b) Use a calculator to estimate $U'(8)$.

[2 points]

(2c) Interpret the sign and units.

[2 points]

(3a) Problem 3: Core Transcendental Derivatives. A population model is $P(t) = 400e^{0.18t}$, but this topic has not yet introduced the Chain Rule. A numerical calculator is therefore used directly to estimate rates.

Use a symmetric difference with $h = 0.001$ to estimate $P'(5)$.

[2 points]

(3b) Interpret the estimate.

[2 points]

(3c) Explain why the symbolic derivative requires a later rule beyond the four basic formulas in this lesson. [2 points]

(4a) Problem 4: Tangent, Cotangent, Secant, and Cosecant. A calculator is permitted for $g(x) = x \csc x$ at $x = 1.2$ radians.

Find $g'(x)$ symbolically.

[2 points]

(4b) Use a calculator to estimate $g'(1.2)$.

[2 points]

(4c) Explain why radians must be used for the standard derivative formula.

[2 points]

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